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Machine Learning Assignment - 35

- 1) Define Machine Learning in your own words. Explain how it is different from traditional rule-based programming with a real-world example.

Ans:

Machine Learning is a branch of Artificial Intelligence that enables a computer to learn from historical data and improve its performance without being explicitly programmed for every task. Instead of following fixed rules, it identifies patterns in data and uses those patterns to make predictions or decisions. As more data becomes available, the model can improve its accuracy over time.

The main difference is that in traditional programming, the programmer writes all the rules manually. The computer simply follows those rules to produce an output. If the rules change, the program must also be updated manually. In Machine Learning, we don't write all the rule. Instead, we provide the computer with training data, it learns the patterns on its own. The more relevant data it receives, the better its predictions becomes. In a Machine Learning, we train the model using thousands of spam and a

- 2) What are Independent Variables and Dependent Variables in Machine Learning? Using the Ball classification case study, clearly identify:

- Independent variables
- Dependent variables

Ans : In Machine Learning, the data is divided into two parts:

- Independent Variable (X): These are the input values or features that are used to make a prediction.
- Dependent Variable (Y): This is the output or target value that the model tries to predict.

Ball Classification Case Study -

Suppose we want to classify different types of balls, such as Cricket Ball, Tennis Ball.

Independent Variable (Features):

- Color
- Texture (Smooth / Rough)
- Weight

These are the characteristics of the ball that help the model identify it.

Dependent Variable (Label):

- Type of Ball
 - Cricket Ball
 - Tennis Ball

3) Explain the difference between Features and Labels.
Are features always numeric? Justify Justify your answer.

parts :

Ans :

Features :

Features are the characteristics or properties of the data. They are used as inputs to train the model.
for ex :- Height, weight, Age, color, size.

Labels :

Labels are the correct answers or outputs. They tell the model what it should learn to predict.

for ex :-

In ball classification :

- Feature : Color, size, weight
- Label : Football, Tennis ball, Cricket ball

Difference

Features

Labels

Input to the model

Output from the model

Describe the object

Identify the object

Independent variables

Dependent variable

Features are not always numeric. They can be :
Numeric, Categorical, Text, Image, Audio.

Before training, Machine Learning algorithms usually convert non-numeric features into numeric values using encoding techniques.

4) What is Supervised Machine Learning? Why is labeled data mandatory in supervised learning?

Ans: Supervised Machine Learning is a type of Machine Learning in which the model is trained using labeled data. This means that for every input, the correct output is already known. The model learns the relationship between the input and the output so that it can predict the correct output for new, unseen data.

Example -

Training Data :

Size	Weight	Label
Small	60g	Tennis Ball
Large	620g	Basket Ball
Medium	450g	Football

The model learns from these examples.

Labeled data acts like a teacher/trainer.

Without labels :

- The model does not know the correct answer
- It ~~is~~ cannot compare its prediction with the actual result.
- Therefore, it cannot learn or improve.

5) Explain why we cannot train and test a model using the same dataset. What problem may occur if we do so?

Ans: we should not use the same dataset for both training and testing because the model may simply memorize the training data instead of learning the underlying patterns.

As a result, It performs very well on the training data. But it performs poorly on new, unseen data.

This gives a false impression that the model is highly accurate.

The main problem is called Overfitting.

Overfitting means the model memorizes the training data instead of learning general patterns. Because of this, it cannot make accurate predictions on new data.

Example -

Suppose a teacher gives students the same questions in the practice test and the final exam. Students may score 100% because they memorized the answers. But if the final exam contains new questions, they may perform poorly. Similarly, a Machine Learning model must be tested on new data to measure its real performance.

Q) What is the purpose of splitting a dataset into training and testing sets? What is the typical ratio used in industry?

Ans: We split a dataset into training and testing sets so that we can both train the model and evaluate how well it performs on new, unseen data.

- The training set is used to teach the model by helping it learn patterns from the data.
- The testing set is used to check whether the model can make accurate predictions on data it has never seen before.

If we use the same data for both training and testing, the model may simply memorize the data instead of learning, which leads to overfitting.

The most commonly used split in the industry is 80% training and 20% testing (80:20). Depending on the dataset, other ratios such as 70:30 or 90:10 may also be used.

7) Explain the meaning of the following variables:

- X-train:

X represents the features (independent variables).

X-train contains the training features that are used to teach the model.

Example:-

Color, Size, Weight of different balls used for training.

- X-test:

X-test contains the testing features. The model uses these features to make predictions after training.

Example:-

The model receives the Size and Weight of a new ball and predicts its type.

- Y-train:

Y-train represents the labels or target values (dependent variable). Y-train contains the correct answers for the training data. The model compares its predictions with these labels and learns from its mistakes.

Example:-

Football, Tennis Ball, Cricket Ball.

• Y-test :

Y-test contains the actual labels of the testing data. It is used to compare with the model's predictions and calculate accuracy.

Simple Example -

Suppose we have this dataset :

Size	Weight	Ball Type
Small	60 g	Tennis Ball
Medium	450 g	Football
Large	620 g	Basket ball

After splitting :

- X-train $\rightarrow$ Size, Weight (training data)
- Y-train $\rightarrow$ Ball Type (training labels)
- X-test $\rightarrow$ Size, Weight (testing data)
- Y-test $\rightarrow$ Ball Type (testing labels)

8) What is the role of the parameter `test_size = 0.2` in `train_test_split()`? What happens if we change it to 0.5?

Ans :

The `test_size` parameter decides how much of the dataset will be used for testing.

- If `test_size = 0.2`, then 20% of the data is used for testing, and the remaining 80% is used for training.
- This is the most commonly used split because it provides enough data for both training and testing.

If we change it to `test_size = 0.5` :

- 50% of the data will be used for training.
- 50% of the data will be used for testing.

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This means the model gets less data to learn from. If the dataset is small, the model may not learn enough patterns, which can reduce its performance.

Example -

Suppose we have 100 records

- test_size = 0.2
 - Training = 80 records
 - Testing = 20 records

9) What is the purpose of random_state = 42 in train_test_split()? What will happen if we do not provide it?

Ans: random_state is used to make the dataset shuffle.

When we split a dataset into training and testing sets, the data is selected randomly. By setting random_state = 42, we ensure that the same records are selected for training and testing every time the program runs.

The number 42 has no special meaning - it is simply a commonly used fixed value. We can use any integer, such as 10, 25 or 100.

If we do not provide random_state, the dataset will be split differently each time the code is executed. As a result, the model's accuracy may change slightly from one run to another.

Example -

Suppose we have 100 records

- With random_state = 42:
 - Every run gives the same 80 training records & the same 20 testing records.
- Without random_state:
 - The training & testing records may be different each time, so the results may vary.

10) Explain step-by-step what happens internally when we execute:
`model.fit(X_train, Y_train)`

Ans: When we execute `model.fit(X_train, Y_train)`, the model starts learning from the training data. The process happens in the following steps:

Step 1: Read the training data

The model reads the input features (`X_train`) and the correct outputs (`Y_train`)

Step 2: Find patterns

The model analyzes the relationship between the input features & the target values. It tries to understand how the inputs affects the output.

Step 3: Learn the parameters

The algorithm adjusts its internal parameters (such as weights, thresholds, or decision rules, depending on the algorithm) to best match the training data.

Step 4: Minimize errors

The model compares its predictions with the actual labels (`Y_train`) and updates its parameters to reduce prediction errors.

Step 5: Create a trained model

After learning from all the training data the model stores the learned patterns. This trained model is then ready to make predictions on new, unseen data.

Example -
Suppose we are training a model to classify balls.

X_train:

- Weight
- Color
- Texture

Y_train:

- Tennis Ball
- Cricket Ball

After calling:

`model.fit(X_train, Y_train)`

the model learns the relationship betⁿ the features (weight, color, texture) and the ball type. Later, when a new ball is given, it can predict its type using:

`model.predict(X_test)`